Longitudinal Propensity Score Analysis: Method Comparison

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1 Objective

This report compares treatment effect estimates across various analytic strategies and scenarios of confounding (
rho_{uv}). We explore variations of the populations (Intention-to-Treat (ITT), Per-Protocol (PP), and Propensity Score Matching (PSM)) across both visit-level and patient-level adherence.

1.1 Outcome Type Interpretations

The simulation framework dynamically adapts to the selected outcome type (continuous). Depending on the configuration, the results in this report should be interpreted as follows:

- **Continuous Outcomes:** Treatment effects are modeled using standard linear/mixed-effects models. The `Estimate` (and Bias) represents the absolute difference in expected means between the treatments.
- **Binary Outcomes:** Treatment effects are modeled using logistic regression with binomial link functions. The `Estimate` (and Bias) is interpreted on the log-odds scale, and the estimates represent log-odds ratios.

2 Representative Scenario (Rho=0.5, Adherence ~75%)

2.1 Patient Level Comparisons

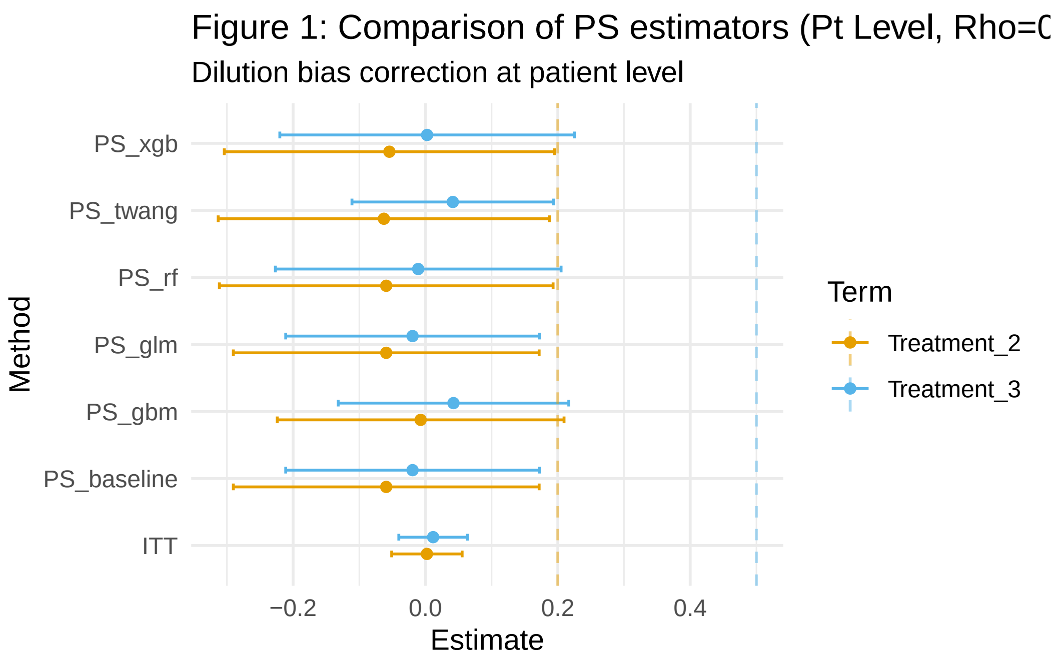


Figure 1

2.2 Observation Level Comparisons

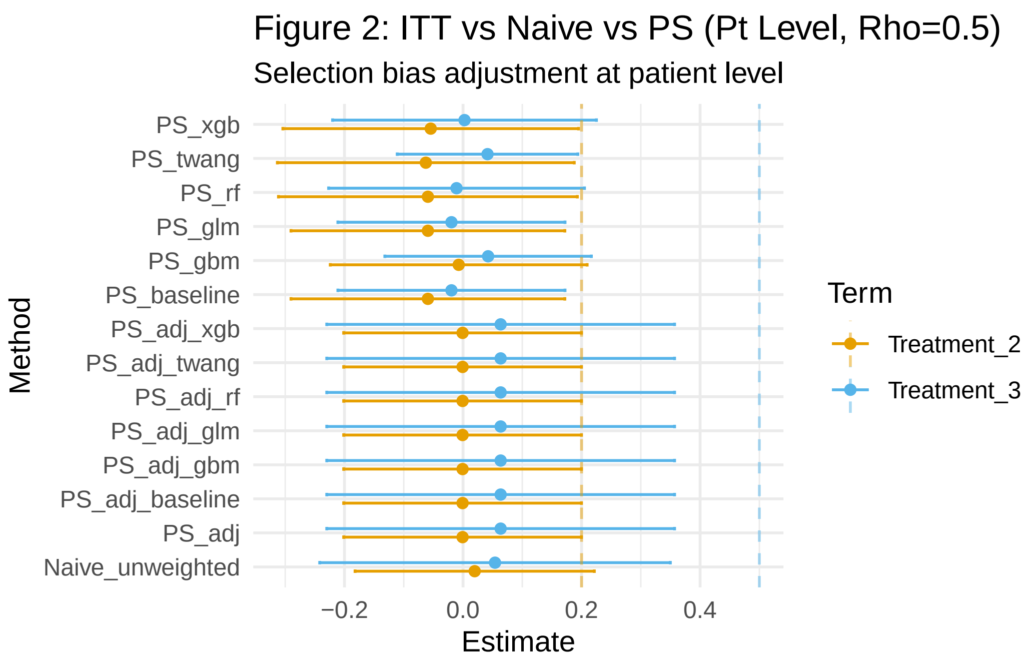


Figure 2

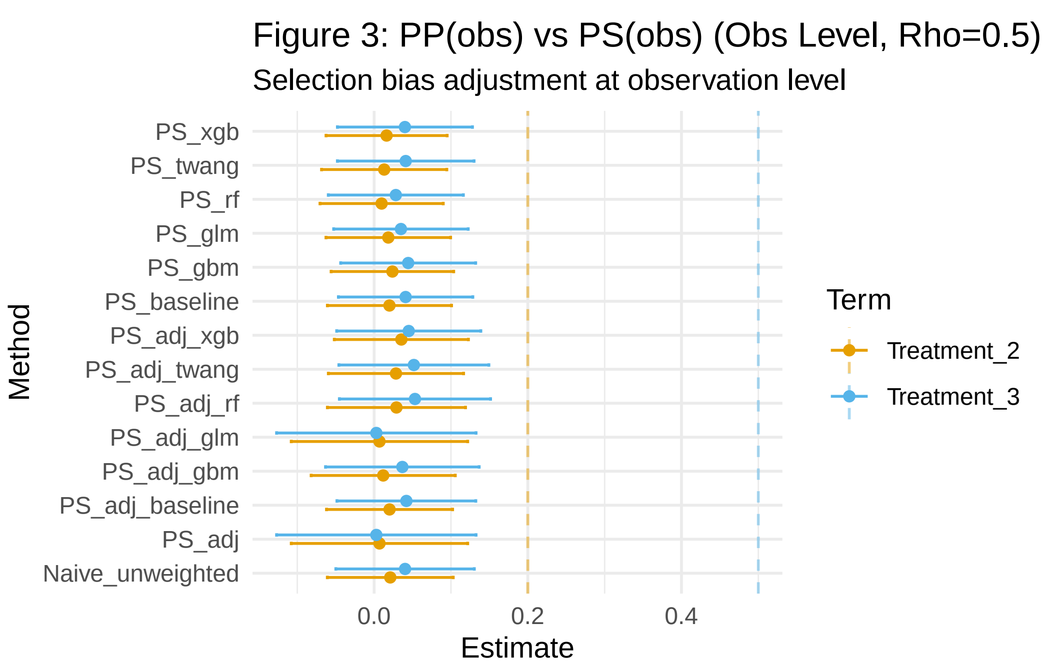


Figure 3

3 Method Comparison (Pooled)

Below we compare the estimated treatment effects across all scenarios.

Table 1: Averaged Treatment Effects Across Scenarios (including Twang)

Level	Method	Term	Mean_Estimate	SD_Estimate
obs	ITT	Treatment_2	-0.004	0.027
obs	Naive_unweighted	Treatment_2	0.001	0.083
obs	PS_adj	Treatment_2	0.001	0.107
obs	PS_adj_baseline	Treatment_2	0.008	0.104
obs	PS_adj_gbm	Treatment_2	0.007	0.105
obs	PS_adj_glm	Treatment_2	0.001	0.107
obs	PS_adj_rf	Treatment_2	0.008	0.104
obs	PS_adj_twang	Treatment_2	0.008	0.104
obs	PS_adj_xgb	Treatment_2	0.007	0.104
obs	PS_baseline	Treatment_2	-0.004	0.079
obs	PS_gbm	Treatment_2	-0.005	0.087
obs	PS_glm	Treatment_2	0.000	0.081
obs	PS_rf	Treatment_2	-0.024	0.121
obs	PS_twang	Treatment_2	-0.005	0.094
obs	PS_xgb	Treatment_2	-0.006	0.084
pt	ITT	Treatment_2	NA	NA
pt	Naive_unweighted	Treatment_2	NA	NA
pt	PS_adj	Treatment_2	NA	NA
pt	PS_adj_baseline	Treatment_2	NA	NA
pt	PS_adj_gbm	Treatment_2	NA	NA
pt	PS_adj_glm	Treatment_2	NA	NA
pt	PS_adj_rf	Treatment_2	NA	NA
pt	PS_adj_twang	Treatment_2	NA	NA
pt	PS_adj_xgb	Treatment_2	NA	NA
pt	PS_baseline	Treatment_2	NA	NA
pt	PS_gbm	Treatment_2	NA	NA
pt	PS_glm	Treatment_2	NA	NA
pt	PS_rf	Treatment_2	NA	NA
pt	PS_twang	Treatment_2	NA	NA
pt	PS_xgb	Treatment_2	NA	NA
obs	ITT	Treatment_3	-0.003	0.027
obs	Naive_unweighted	Treatment_3	-0.002	0.081
obs	PS_adj	Treatment_3	-0.006	0.119
obs	PS_adj_baseline	Treatment_3	0.001	0.116
obs	PS_adj_gbm	Treatment_3	-0.002	0.117
obs	PS_adj_glm	Treatment_3	-0.006	0.119
obs	PS_adj_rf	Treatment_3	0.002	0.116
obs	PS_adj_twang	Treatment_3	0.001	0.117
obs	PS_adj_xgb	Treatment_3	-0.001	0.117
obs	PS_baseline	Treatment_3	0.001	0.086

Level	Method	Term	Mean_Estimate	SD_Estimate
obs	PS_gbm	Treatment_3	-0.002	0.084
obs	PS_glm	Treatment_3	0.001	0.089
obs	PS_rf	Treatment_3	0.002	0.094
obs	PS_twang	Treatment_3	0.004	0.080
obs	PS_xgb	Treatment_3	0.000	0.084
pt	ITT	Treatment_3	NA	NA
pt	Naive_unweighted	Treatment_3	NA	NA
pt	PS_adj	Treatment_3	NA	NA
pt	PS_adj_baseline	Treatment_3	NA	NA
pt	PS_adj_gbm	Treatment_3	NA	NA
pt	PS_adj_glm	Treatment_3	NA	NA
pt	PS_adj_rf	Treatment_3	NA	NA
pt	PS_adj_twang	Treatment_3	NA	NA
pt	PS_adj_xgb	Treatment_3	NA	NA
pt	PS_baseline	Treatment_3	NA	NA
pt	PS_gbm	Treatment_3	NA	NA
pt	PS_glm	Treatment_3	NA	NA
pt	PS_rf	Treatment_3	NA	NA
pt	PS_twang	Treatment_3	NA	NA
pt	PS_xgb	Treatment_3	NA	NA

4 Visualizing Bias and Coverage

4.1 Bias vs Confounding Strength (Rho Sweep)

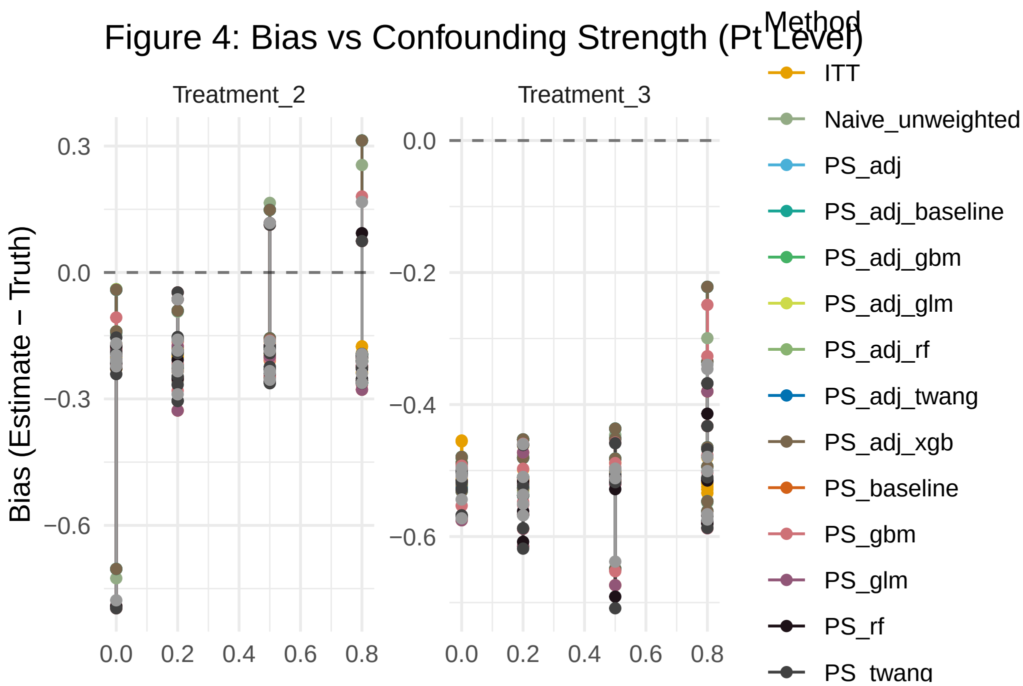


Figure 4

4.2 Bias vs Observed Adherence Rate (Adherence Sweep)

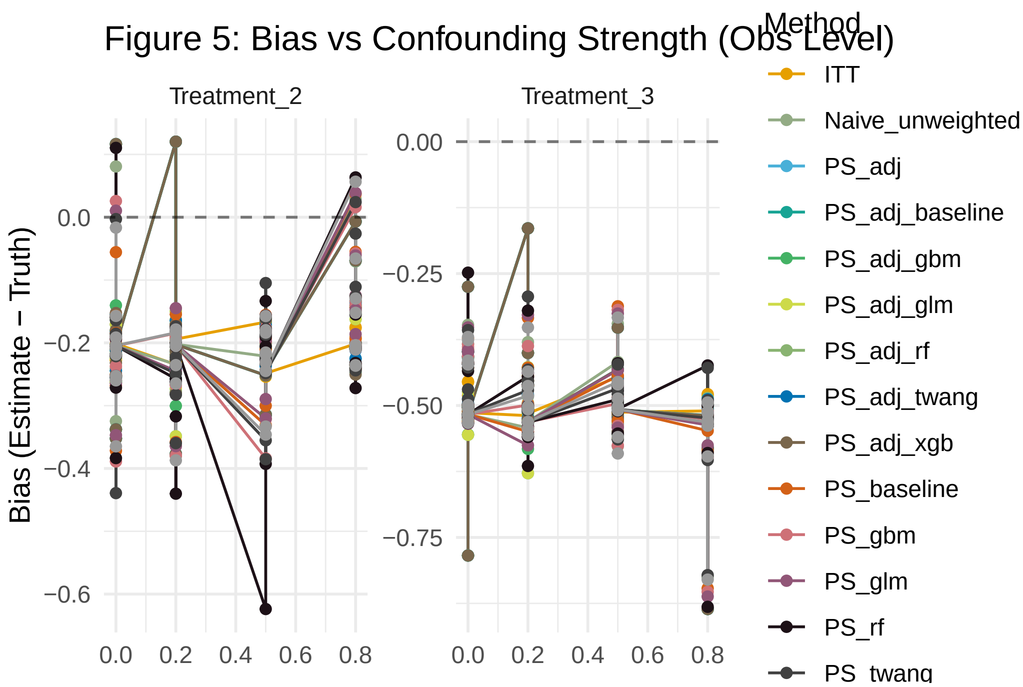


Figure 5

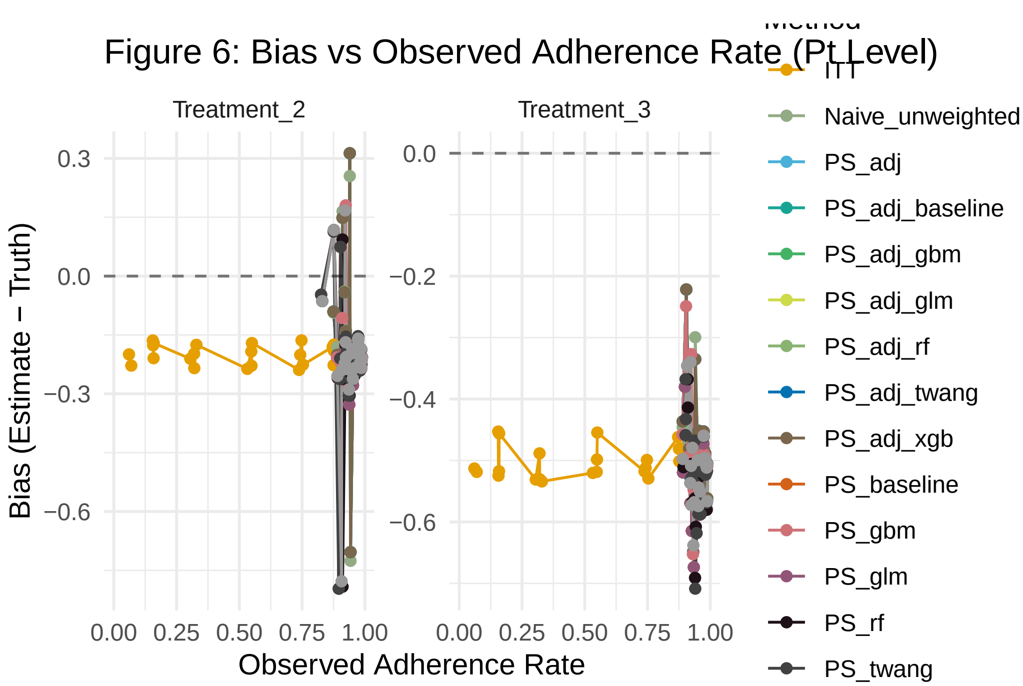


Figure 6

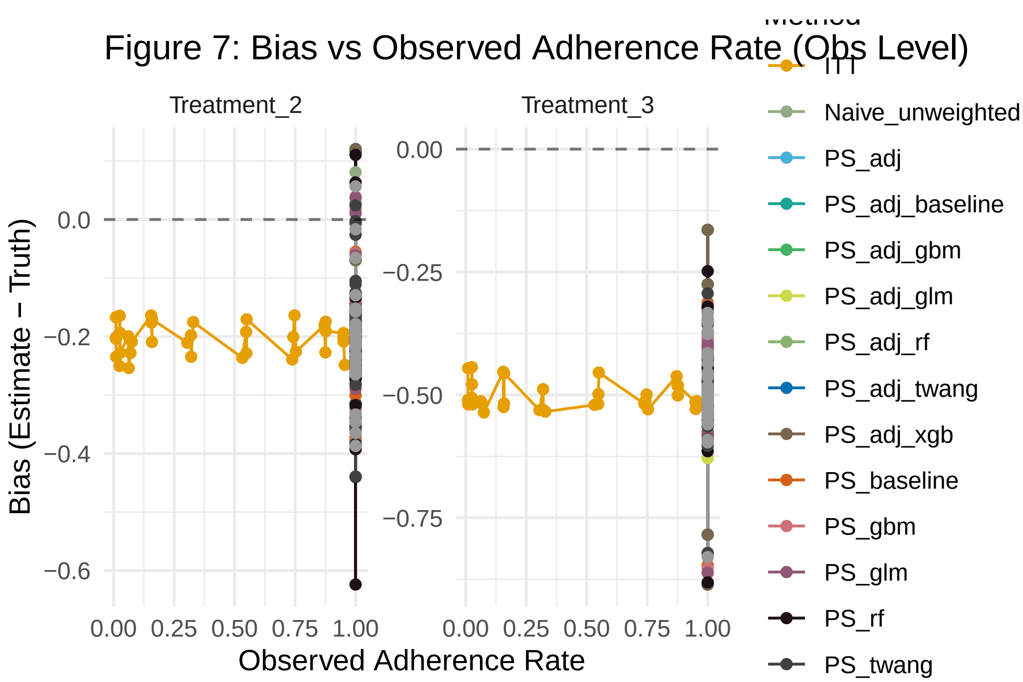
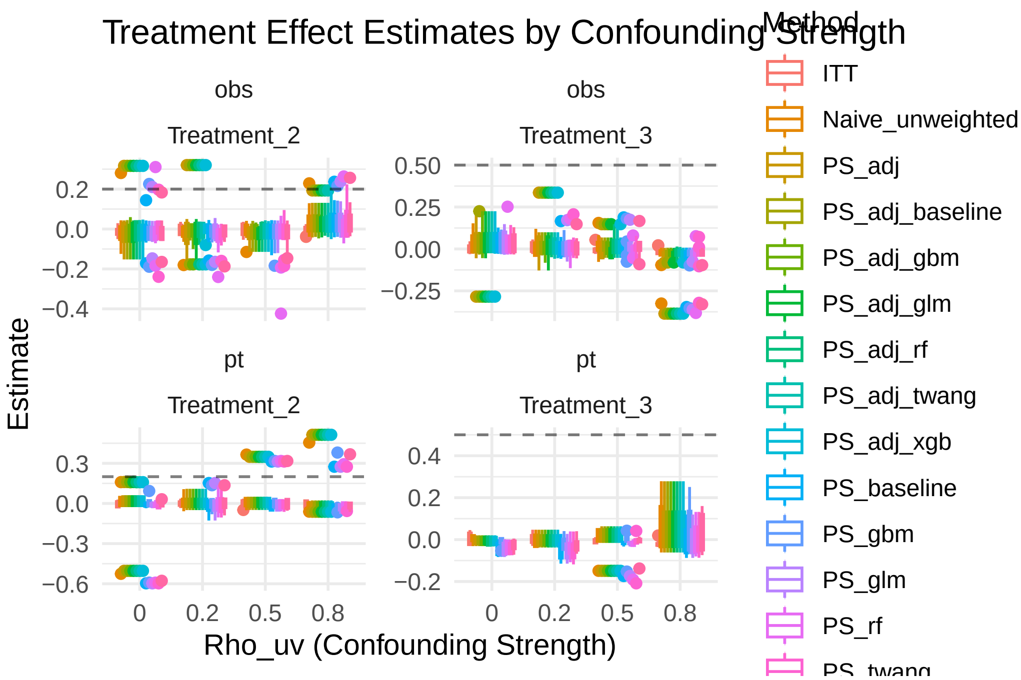


Figure 7: Bias vs Observed Adherence Rate (Obs Level)
Treatment_2
Treatment_3
Bias (Estimate − Truth)
Observed Adherence Rate
0.0
−0.2
−0.4
−0.6
0.00
−0.25
−0.50
−0.75
0.00 0.25 0.50 0.75 1.00
0.00 0.25 0.50 0.75 1.00
Method
ITT
Naive_unweighted
PS_adj
PS_adj_baseline
PS_adj_gbm
PS_adj_glm
PS_adj_rf
PS_adj_twang
PS_adj_xgb
PS_baseline
PS_gbm
PS_glm
PS_rf
PS_twang

Figure 7

5 Longitudinal Estimates Distribution

Grouping by ρ_{uv} to see how estimates distribute relative to truth.



6 Discussion

Machine learning PS methods (GBM, XGBoost, RF) along with `twang` are evaluated against parametric GLM and naive ITT/PP approaches. As
rho_{uv} increases, the selection bias in PP approaches typically grows, while PS methods attempt to recover the true effect. The analysis confirms that understanding the adherence-outcome correlation is critical in deciding the appropriate analytic approach, especially in the context of the specific outcome type evaluated.